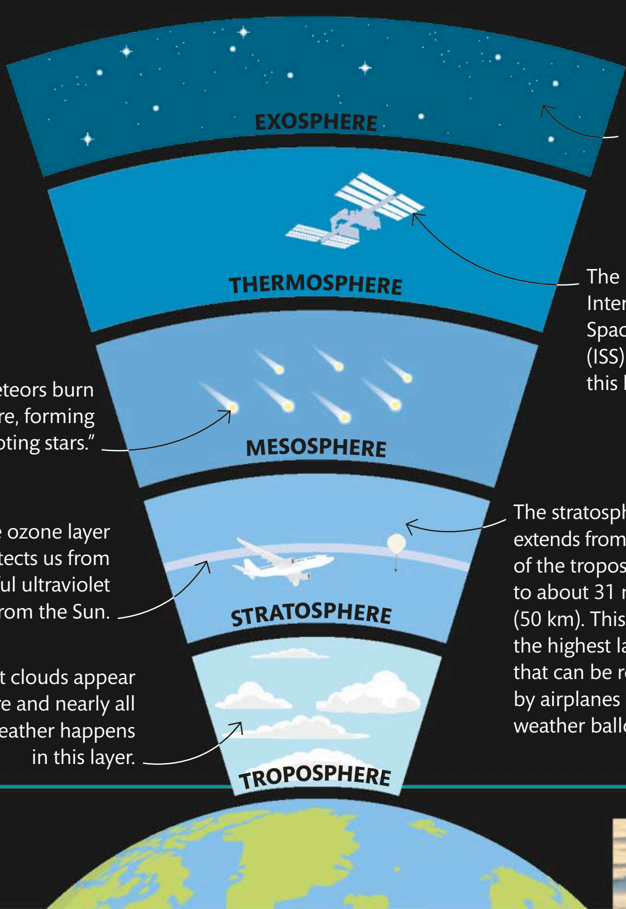
STUDYING THE ATMOSPHERE

WEATHER BALLOONS

Earth is surrounded by a layer of gases called the atmosphere. The two main gases are nitrogen and oxygen, with small amounts of others. The atmosphere supports life on Earth by absorbing harmful radiation from the Sun, trapping heat, and generating pressure, which allows liquid water to exist on Earth. Understanding the atmosphere can tell us about weather patterns, pollution, climate change, and more. One way to study it is using weather balloons, which can carry scientific instruments high up into the atmosphere and send back data to scientists on Earth.

EARTH'S ATMOSPHERE

The atmosphere is made up of distinct layers that become less dense as the distance from Earth's surface increases, until the outer layer merges with space. The closest layer to Earth is the troposphere, which contains the air we breathe.



Most satellites orbit Earth in this layer.

The International Space Station (ISS) orbits in this layer.

Most meteors burn up here, forming "shooting stars."

The ozone layer protects us from harmful ultraviolet rays from the Sun.

The stratosphere extends from the top of the troposphere to about 31 miles (50 km). This is the highest layer that can be reached by airplanes and weather balloons.

Most clouds appear here and nearly all weather happens in this layer.

Gathering data

Weather balloons carry scientific instruments that can measure atmospheric pressure, humidity, and temperature. They can also identify the gases present at different heights in the atmosphere. This time-lapse photo shows a balloon being released by scientists in Antarctica. It is carrying an instrument to monitor ozone levels.

The instruments carried by the balloon can be tracked by radar or GPS.